

Joseph Lodato

jalodato@syr.edu | (207) 841-2535 | [josephlodato.github.io](https://github.com/josephlodato) | [linkedin.com/in/joseph-lodato/](https://www.linkedin.com/in/joseph-lodato/)

EDUCATION

Syracuse University – M.S. Computer Engineering

Anticipated May 2027

Syracuse University – B.S. Computer Engineering, Minor: Physics

Magna Cum Laude, GPA 3.60 | May 2026

Relevant Coursework: Semiconductor Fabrication, System-on-Chip Design, Advanced Computer Architecture, Signals & Systems Analysis, Embedded Systems & Microcontroller Lab, Computational Physics, VLSI Design, Computer & Network Security, Database Management System

TECHNICAL SKILLS

PCB Design & Hardware Engineering: Schematic capture & multi-layer PCB layout (KiCad), SPICE simulation, DC-DC/switching regulator design, gate drivers & MOSFETs, high-voltage systems, battery management systems (BMS), DFM/DFA, signal integrity

Embedded Systems & Firmware: Microcontroller firmware (PIC16, nRF52, Teensy 4.0), I2C/SPI/UART, real-time embedded control

Test & Measurement: Oscilloscopes, multimeters, bench power supplies & electronic loads, hardware bring-up & debug

Languages & Tools: C/C++, C#, Java, Python, Assembly, Git, KiCad, Fusion 360, OpenCV

Fabrication & Platforms: Machine shop (lathe, mill, 3D printing), Windows, Linux

HARDWARE & PCB PROJECTS

120V, 10A DC-DC Supercapacitor Charger – Citrus Racing EV Team

In Progress

- Designing a synchronous half-bridge DC-DC converter (IRS2008S gate driver, RCX200N20 MOSFETs, 100uH inductor, PIC16F1769 firmware control) rated for 120VDC / 10A continuous to step up a 72V battery array to charge a 97V nominal supercapacitor bank.
- Sized the supercap bank to absorb acceleration inrush current beyond the ~10A limit of the batteries' internal BMS units, with a blocking diode preventing regen-braking energy from back-feeding into the array; validating on perfboard ahead of PCB layout.

'IRIS' Early Warning Wearable Weather System – Custom PCB

Personal PCB Project

- Designing a wearable alpine weather system providing early warning for incoming storms for hikers and climbers. Used KiCad integrating 8+ sensors (lightning detection, GPS, barometric, UV, IMU) on an I²C bus with an nRF52840 MCU and an SPI e-ink display.
- Resolved power-rail and load-switching issues for high-current peripherals (haptic motor, addressable LED array) through iterative schematic review and testing.

'Apogee' Fully Custom Mechanical Keyboard – 2 Layer PCB

Personal PCB Project

- Designed a fully custom 2-Layer PCB. With a nRF52840 MCU and MCP23017 I²C GPIO expander for key matrix scanning, balancing pin budget against Bluetooth and peripheral I/O needs. Revised firmware to optimize for latency, resulting in 2.8 ms end-to-end latency.
- Integrated BLE connectivity, OLED status display, rotary encoders, and a TRRS expansion port; tuned power budget for ~5hr battery life on ZMK firmware.

'Hockey Bot' - Undergraduate Capstone Project

4-Person Team

- Built an autonomous air hockey robot that plays a human in a game of air hockey. Tracks a live puck with an OpenCV pipeline and moves to intercept and return it in real time, with three selectable difficulty levels.
- Designed and built the CoreXY belt driven gantry mechanism and motor driver communication, converting USB serial data from a vision processing host computer into real-time NEMA-17 stepper commands via Teensy 4.0.

LEADERSHIP

Formula SAE – Citrus Racing, Syracuse University

Oct 2022 – Present

Lead of EV Development (Current)

- Directing design and fabrication of a high-voltage battery pack (currently 72V, scaling to 144V for the first full vehicle build) from commercially available 12/24V LiFePO₄ modules with integrated cell-level BMS, plus a top-level pack BMS, motor control, and vehicle safety systems for the team's first electric powertrain.
- Leading the supercapacitor charging subsystem design (see Projects) as part of the vehicle's energy management architecture.

Prior Roles: Chief Engineer, President, Lead of Electronics

- As Chief Engineer, led the 2026 vehicle development cycle to the team's highest Formula SAE score in the 2000s era among 120 competing teams.
- As President, managed a \$50,000 annual budget and grew active membership from ~5 to 40+ through sponsorship and recruitment efforts.
- As Lead of Electronics, designed all vehicle electrical systems including wiring harnesses, ECU control software, and data acquisition.

WORK EXPERIENCE

AEC Engineering – Control Panel Technician

May 2025 – Sep 2025

- Fabricated and verified industrial control panels to UL508A standards, including ring-outs, functional testing, and PLC logic debugging; developed a weight-based inventory tracking framework for shop components.

Syracuse University Student Machine Shop – Student Machinist

Aug 2023 – Present

- Support students across lathes, mills, and 3D printers for coursework and independent projects, ensuring safe and proper equipment operation.